

Ali Azhar Dewali

Biomedical Engineering Student — Brain-Computer Interfaces & Embedded Systems

ali.azhar.dewali@gmail.com · WhatsApp +964 750 792 4022 · Instagram @alii.azhar
aliazhardewali-pixel.github.io · github.com/aliazhardewali-pixel · Duhok, Kurdistan Region, Iraq



EDUCATION

B.Sc. Biomedical Engineering — Double Degree · University of Duhok (UoD) & TH Mittelhessen (THM), Germany

4th-year student · degree taught fully in English · expected graduation 2027. Six-month biomedical engineering internship in Germany planned for 2027 as part of the program.

Secondary education: Werar High School, 2023.

EXPERIENCE

Biomedical Engineering Intern — Dr. Ghazi Hospital, Duhok · Jul–Sep 2025

- Operation, repair, calibration and installation of clinical medical devices; supervised by Eng. Mohammed Mizori.
- Maintenance and inspection of high-capacity oxygen and CO₂ generation plants; troubleshooting of hospital-wide medical vacuum systems used in critical care.
- Recognized for professional conduct and commitment to hospital safety standards.

TECHNICAL PROJECTS

- **EEG-Controlled Aerodynamic Levitation (flagship)**. NeuroSky EEG headset → Python engine measuring focus in real time → ESP32-driven fan levitating a ping-pong ball. Full pipeline: acquisition, signal parsing, motor control. Presented at the 13th Engineering Design Day, UoD. Code: github.com/aliazhardewali-pixel/EEG-Levitation-BCI
- **ECG Monitor Design**. Functional simplified ECG using three bio-potential electrodes to capture and display the heart signal.
- **Microcontroller Motor Control**. DC motor speed control on Arduino Uno/Nano and ESP32 with closed-loop PID and live LCD feedback.
- **Multibody System Simulations**. Mechanical joint and spring-system modeling and analysis in Dymola (Modelica).
- **Patient Intake Sheet Tool**. Browser tool for physicians — fill in patient data, print a clean one-page intake sheet.
- **C++ Calculator**. Console calculator with input validation and clean control flow.

SKILLS

Programming: C++, Python · **Engineering software:** MATLAB, Dymola, PlatformIO, Arduino IDE, Visual Studio

Data & productivity: Power BI, MS Office · **Hardware:** ESP32, Arduino, circuit design

CERTIFICATIONS & TRAINING

- Certificate of Participation — 13th Engineering Design Day, University of Duhok.
- Leadership Training (22h) — Kurdistan Students Development Organization, 2023.
- Rules of Success Course (7h) — Infinity Organization, 2020 · English Language Training — Da Vinci Institute, 2020.

LANGUAGES

Kurdish (Badini) — native · Arabic — fluent · English — fluent (degree taught in English) · currently adding German & Sorani